CS 5316– NLP Theory and Applications
Course Project

Introduction
Natural language processing (NLP) has generated immense interest in the industry and the academia in recent years. Advances in machine learning have made NLP tasks highly effective and many NLP applications provide value to personal and business users.
The primary purpose of the course project is to expose you to current challenges and solutions in NLP. It is expected to build confidence and skills in re-implementing recent research and developing value addition applications.

Description
Broadly, the projects can fall into one of the following two categories: (1) Develop an NLP application or solution that is novel and/or valuable; (2) Re-implement a recently published work generating and validating results on similar datasets. Most of the projects are expected to fall in category 2. Nonetheless, the distinction between the two categories can be blurred in some cases so you should familiarize your self with the description of both categories.
Category 1: You are required to develop a prototype of a application or solution involving NLP. This project should satisfy the following conditions:

1. The idea should be concrete with easily understandable novelty and/or value
2. Sufficient data is available or is readily generatable for the task
3. Quantifiable methods are available for evaluating the output of the task
4. The task should have reasonable complexity involving multiple models and/or datasets
5. The project team has reasonable understanding of the task's domain area

One way to identify interesting applications is to address challenges and opportunities in the local social, economic, cultural, linguistic, etc contexts. Brainstorm with your favorite LLMs, amongst yourselves, Tas, and instructor to formalize your project. A working solution is required at the end of the project. The focus should not be on UI elements but NLP and its evaluation.
Category 2: You are required to search and select a recent research publication for re-implementation. The publication should satisfy the following conditions:

1. The publication should be recent, i.e., 2022 and later.
2. The paper should be published at a reputable venue, and should have at least 8 pages.
3. The paper should involve significant NLP (primary focus on other areas, e.g., computer vision are not allowed).
4. The dataset used in the work should be readily available.

Some tips for searching for papers:

1. Engage your favorite LLM on recent trends and challenges in NLP. Direct the discussion using focused prompts that emphasize topics of interest (e.g., see below) and quality of work.
2. Search on research portals: Google Scholar, ACL Anthology, IEEE Xplorer, ACM Digital Library, etc.
3. Browse through listing of relevant conferences and journals: ACL, NAACL, EMNLP, COLING, LREC, EACL, ICLR, Computational Linguistics, Natural Language Engineering, Transactions of ACL, Language Resources and Evaluation, TALLIP, etc.
4. Browse NLP competitions and challenges sites: SemEval, Kaggle, etc.

You may shortlist 3 to 5 papers. Discuss them with the instructor or Tas before finalizing one. Your task will be to study and understand the paper thoroughly. You should then re-implement and evaluate the models/algorithms presented in the paper. If you are re-using code made available by the authors then you should clearly mention this in your presentation/report. Runs experiments on the data used in the paper. Present and discuss the results of your evaluation. Generate and discuss results on alternate datasets.

Team Size
A project team can have a maximum of 4 members.

Procedure and Deadlines
Groups: Submit your group information to the TA. People who have difficulty in forming groups OR do not send in their entries will be assigned to groups made by the TA. (Deadline: October 12)
Initial meeting and proposal: Talk to the instructor or the TAs to discuss your project ideas before finalizing one. You should have at least one meeting with us before your proposed project can be approved. Submit a proposal on the finalized project. Please come prepared for the meeting. (Deadline: October 23)
Progress meeting and intermediate report: update the instructor or the TA on your project status at least once after initial meeting and before the final submission. The meeting will have points so make sure you don't miss out on them. You will also be required to submit a progress report. (deadline: TBA)
Presentation/Viva/Demo: give a 15 -minute presentation/viva/demo summarizing the findings of your project (It may involve a recorded presentation. Deadline: TBA, will be in final week).
Final report: submit a well-written final report that presents the findings of your work. It should include introduction, brief review of related works, overview of the models/algorithms, results of your experiments, and discussions/insights. Greater emphasis should be placed on the experiments and results section. (Deadline: TBA in exam week)

Grading

Initial meeting and proposal	15 points
Progress meetings	15
Presentation/demo/Viva	35 points
Final report	35 points

Project Ideas
You are encouraged to work with **large language models (LLMs), multilingual NLP, low-resource languages, foundation models, agentic AI, retrieval-augmented generation (RAG), instruction tuning, parameter-efficient fine-tuning, bias, robustness, and hallucination**.
A good project should answer a question such as:
Does method X improve NLP performance under condition Y, and why?
or
How does an NLP/foundation model behave differently under condition X?
Projects may be experimental, comparative, or application-oriented, but they must contain a meaningful **research component**.

)Broad topics of interest in our research group are:

1. Urdu, Roman Urdu, and other low-resource language processing (any aspect, task, or application)
2. Study of prompting in NLP especially for local languages
3. Exploiting Post-tuning such as instruction tuning, preference optimization, reinforcement learning, constitutional AI, etc for NLP solutions
4. Agentic AI frameworks and solutions for NLP challenges.
5. Small language models vs. LLMs especially for low-resource NLP
6. Characterizing LLM as judge and LLMs for reasoning

Innovative IR, RAG, and QA solutions

7. Applications of NLP in the medical domain
8. Characterizing and mitigating LLM harm (hallucinations, bias, privacy, security, etc)
9. Explainability in NLP